

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Dept. of E&ECE

2. Subject Name: Embedded Systems Laboratory L-T-P: 0-0-3 Credits: 2

3. Pre-requisites: -

4. Syllabus and reference books:

Write a program in assembly language of 8051 to find the second largest in an array of 10 numbers stored from a memory location; Design a traffic light controller; Design a digital clock; Show the operation of interrupt facility of ARM board; Communicate between the two SPI ports of ARM to exchange data bytes; Interface an ADC to the ARM processor. Use the interface to measure the frequency of slow running sine wave; Interface a stepper motor with the ARM processor.

Reference Books:

N/A

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Write a program in assembly language of 8051 to find the second largest in an array of 10 numbers stored from a memory location.	3
2.	Design a traffic light controller.	6
3.	Design a digital clock.	6
4.	Show the operation of interrupt facility of ARM board.	3
5.	Communicate between the two SPI ports of ARM to exchange data bytes.	6
6.	Interface an ADC to the ARM processor. Use the interface to measure the frequency of slow running sine wave.	6
7.	Interface a stepper motor with the ARM processor.	6
Total number of hours		36